

1. Example of Bayes Rule

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Problem Example (Medical Diagnosis)

Suppose:

- 1% of people have a disease
- 99% of people do not have the disease
- Test accuracy:
 - If a person **has the disease**, the test is positive **95%** of the time
 - If a person **does not have the disease**, the test is positive **5%** of the time

We want to calculate:

What is the probability that a person actually has the disease if the test result is positive?

Step 1: Given Probabilities

Probability	Value
P(Disease)	0.01
P(No Disease)	0.99
P(Positive Disease)	0.95
P(Positive No Disease)	0.05

Step 2: Apply Bayes Rule

Posterior probability:

P(Disease | Positive)

$$= (0.95 \times 0.01) / [(0.95 \times 0.01) + (0.05 \times 0.99)]$$

$$= 0.0095 / (0.0095 + 0.0495)$$

$$= 0.0095 / 0.059$$

$$\approx \mathbf{0.161}$$

Final Result

Probability that the person actually has the disease $\approx 16.1\%$

Even though the test is accurate, the probability is still low because the disease is **rare**.

Teaching Insight

This example shows how **Bayes Rule updates belief when new evidence is observed**.

2. Example of Bayesian Belief Network

Definition

A **Bayesian Belief Network** is a graphical model that represents **probabilistic relationships between variables**.

Simple Example (Weather System)

Variables:

- Rain
- Sprinkler
- Wet Grass

Relationship:

Rain $\rightarrow$ Wet Grass

Sprinkler $\rightarrow$ Wet Grass

Meaning:

- If it rains, grass becomes wet
- If sprinkler is on, grass becomes wet

Structure

Rain $\rightarrow$

Wet Grass

Sprinkler $\rightarrow$

Conditional Probability Table Example

Rain Sprinkler Wet Grass Probability

Yes	Yes	0.99
Yes	No	0.90
No	Yes	0.80
No	No	0.01

Reasoning Example

If we observe:

Grass is Wet

We can infer probabilities of causes:

- Maybe it **rained**
- Maybe the **sprinkler was on**

The network helps compute these probabilities using **Bayesian inference**.

Real Applications

Bayesian belief networks are used in:

- medical diagnosis
- fraud detection
- fault diagnosis
- decision support systems

Apriori Algorithm (OS)

Definition

Apriori Algorithm is a fundamental algorithm used for **mining frequent itemsets and generating association rules** from transactional databases.

It works on the **Apriori property**:

If an itemset is frequent, then **all of its subsets must also be frequent**.

This property helps **reduce the search space** when finding frequent patterns.

Apriori Algorithm Steps

Algorithm: Apriori

Input:

- Transaction database **D**
- Minimum support threshold **minsup**

Output:

- All frequent itemsets

Procedure

1. Find frequent **1-itemsets (L₁)** from the database.
2. Use **L(k-1)** to generate candidate **k-itemsets (C_k)**.
3. Scan database to calculate support for candidates.
4. Remove itemsets with support < **minsup**.
5. Remaining itemsets become **L_k** (frequent itemsets).
6. Repeat until no new frequent itemsets are found.

Example (Classic Han & Kamber Example)

Transaction Database

TID	Items
T100	I1, I2, I5
T200	I2, I4
T300	I2, I3
T400	I1, I2, I4
T500	I1, I3
T600	I2, I3
T700	I1, I3
T800	I1, I2, I3, I5
T900	I1, I2, I3

Minimum Support Count = 2

Step 1: Generate L1 (Frequent 1-Itemsets)

Item	Support Count
I1	6
I2	7
I3	6
I4	2
I5	2

All items satisfy **minsup = 2**

So,

$L1 = \{I1, I2, I3, I4, I5\}$

Step 2: Generate Candidate 2-Itemsets (C2)

Itemset	Support
I1 I2	4

Itemset	Support
I1 I3	4
I1 I4	1
I1 I5	2
I2 I3	4
I2 I4	2
I2 I5	2
I3 I4	0
I3 I5	1
I4 I5	0

Step 3: Generate L2 (Frequent 2-Itemsets)

Remove itemsets with support < 2

L2 =

Itemset	Support
I1 I2	4
I1 I3	4
I1 I5	2
I2 I3	4
I2 I4	2
I2 I5	2

Step 4: Generate Candidate 3-Itemsets (C3)

Possible combinations:

Itemset	Support
I1 I2 I3	2
I1 I2 I5	2

Step 5: Generate L3 (Frequent 3-Itemsets)

Itemset	Support
I1 I2 I3	2
I1 I2 I5	2

Step 6: Generate C4

Candidate:

I1 I2 I3 I5

But support < minsup → **removed**

So algorithm stops.

Final Frequent Itemsets

L1

L2

L3

Frequent itemsets:

{I1}

{I2}

{I3}

{I4}

{I5}

{I1 I2}

{I1 I3}

{I1 I5}

{I2 I3}

{I2 I4}

{I2 I5}

{I1 I2 I3}

{I1 I2 I5}

Advantages of Apriori

- Simple and easy to understand
- Reduces candidate itemsets using Apriori property
- Widely used in **market basket analysis**

Limitations

- Multiple database scans required
- Generates many candidate itemsets
- Slow for **large datasets**

Information Gain

Definition

Information Gain measures how much **uncertainty (entropy)** is reduced after splitting a dataset based on an attribute.

It is used in **decision tree learning** to select the **best attribute for splitting the data**.

Entropy Formula

$$\text{Entropy}(S) = - \sum_{i=1}^n p_i \log_2(p_i)$$

Where

- (p_i) = probability of class (i)
-

Information Gain Formula

$$\text{Gain}(S,A) = \text{Entropy}(S) - \sum_{v \in \text{Values}(A)} \frac{|S_v|}{|S|} \text{Entropy}(S_v)$$

Where

- **S** = dataset
- **A** = attribute
- **S_v** = subset of S where attribute A has value v

Example Dataset (Play Tennis)

Day	Outlook	Temperature	Humidity	Wind	Play Tennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

Total tuples = **14**

Positive (Yes) = **9**

Negative (No) = **5**

Step 1: Calculate Entropy of Dataset

Entropy(S) $\approx$ 0.94

Step 2: Information Gain for Attribute "Outlook"

Outlook = Sunny

5 tuples $\rightarrow$ (Yes = 2, No = 3)

Entropy $\approx$ 0.97

Outlook = Overcast

4 tuples $\rightarrow$ (Yes = 4, No = 0)

Entropy = 0

Outlook = Rain

5 tuples $\rightarrow$ (Yes = 3, No = 2)

Entropy $\approx$ 0.97

Step 3: Calculate Expected Entropy

E(Outlook) $\approx$ 0.69

Step 4: Information Gain

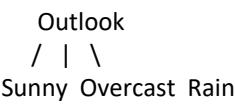
[Gain(S,Outlook) = 0.94 - 0.69]

Gain(S,Outlook) $\approx$ 0.25

Result

Among all attributes, **Outlook** has the **highest information gain**, so it is selected as the **root node of the decision tree**.

Final Decision Tree (First Split)



- Overcast → Play Tennis = Yes
- Sunny and Rain need further splitting

Simple Interpretation for Students

Concept	Meaning
Entropy	Measure of randomness in data
Information Gain	Reduction in entropy after splitting
Best Attribute	Attribute with highest information gain